CBSE Class 12 Pe — Biomechanics and Sports

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark (MCQ/Assertion-Reason/VSA)

Q1. "The more force one exerts on the downward bounce, the higher the ball bounces into the air." Which law is this statement referring to? (A) Newton's First Law (B) Newton's Second Law (C) Newton's Third Law (D) Law of Gravitation [PYQ 2024] | Confidence: Very High

Q2. What type of lever has the load (resistance) located strictly between the fulcrum and the force (effort)? (A) First-class lever (B) Second-class lever (C) Third-class lever (D) Both (A) and (B) [PYQ 2024] | Confidence: Very High

Q3. "The golf ball remains at rest until it is struck by a golf club." This statement indicates: (A) Law of Inertia (B) Law of Acceleration (C) Law of Gravity (D) Law of Reaction [PYQ 2023] | Confidence: Very High

Q4. Which of the following is NOT a factor affecting the trajectory of a projectile? (A) Gravity (B) Angle of release (C) Buoyant force (D) Air resistance [PYQ 2025] | Confidence: Very High

Q5. Given below are two statements labeled Assertion (A) and Reason (R): Assertion (A): When an object or body is moving at a constant velocity, with no change in speed or direction, it is in dynamic equilibrium. Reason (R): A cyclist in motion or the body position maintained by a sprinter on the track are examples of dynamic equilibrium. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. [PYQ 2024] | Confidence: High

Q6. The type of anatomical movement in which the angle between the joint decreases is called: (A) Flexion (B) Extension (C) Gliding (D) Sliding [PYQ 2022] | Confidence: Very High

Q7. Straightening parts of a joint so that the angle increases is medically termed as: (A) Flexion (B) Extension (C) Abduction (D) Adduction [PYQ 2022] | Confidence: Very High

Q8. Moving a body part away from the imaginary mid-line of the body is known as: (A) Flexion (B) Extension (C) Abduction (D) Adduction [PYQ 2022] | Confidence: Very High

Q9. Moving a body part towards the mid-line of the body is known as: (A) Flexion (B) Extension (C) Abduction (D) Adduction [PYQ 2022] | Confidence: Very High

Q10. The path followed by a projectile in the air after being released is called a: (A) Trajectory (B) Velocity (C) Momentum (D) Fulcrum [PYQ 2019] | Confidence: Very High

Q11. Which of the following types of friction primarily helps a skater glide on an ice rink? (A) Rolling friction (B) Sliding friction (C) Static friction (D) Fluid friction [PYQ 2025] | Confidence: High

Q12. Newton’s Second Law of Motion is also universally known as the: (A) Law of Inertia (B) Law of Momentum / Acceleration (C) Law of Reaction (D) Law of Gravitation [PYQ 2024] | Confidence: Very High

Q13. A bicep curl exercise is a classic anatomical example of which class of lever in the human body? (A) First-class lever (B) Second-class lever (C) Third-class lever (D) Fourth-class lever [Practice] | Confidence: Very High

Q14. The optimal angle of release to achieve the maximum horizontal distance for a projectile (like a javelin or shot put) is theoretically: (A) 30° (B) 45° (C) 60° (D) 90° [Practice] | Confidence: Very High

Q15. The invisible downward force that constantly pulls a projectile back towards the earth's surface is: (A) Air Resistance (B) Friction (C) Gravity (D) Centrifugal force [Practice] | Confidence: Very High

Q16. Given below are two statements labeled Assertion (A) and Reason (R): Assertion (A): Friction is often considered a "necessary evil" in sports. Reason (R): While friction can slow down athletes and cause wear and tear, it is absolutely essential for generating grip, such as a runner wearing spiked shoes to prevent slipping. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. [Practice] | Confidence: Very High

Q17. A seesaw in a playground or the nodding of the human head at the neck joint is an example of which type of lever? (A) First-class lever (B) Second-class lever (C) Third-class lever (D) None of the above [Practice] | Confidence: High

Q18. The imaginary central point around which the athlete's entire body weight is evenly balanced in all directions is known as the: (A) Base of support (B) Centre of Gravity (C) Line of gravity (D) Fulcrum [Practice] | Confidence: Very High

Q19. Cartwheel in gymnastics is an example of ___. (A) Static Equilibrium (B) Dynamic Equilibrium (C) Active Equilibrium (D) Passive Equilibrium [PYQ 2023] | Confidence: High

Q20. Define 'Biomechanics' in one sentence. [PYQ 2018] | Confidence: Very High

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2-Mark

Q21. What do you mean by Bio-mechanics? Explain any two points of its importance in sports. [PYQ] | Confidence: Very High

Q22. Differentiate clearly between Static Equilibrium and Dynamic Equilibrium. [Practice] | Confidence: Very High

Q23. Define 'Centre of Gravity'. How does it affect an athlete's stability? [PYQ] | Confidence: High

Q24. State Newton’s First Law of Motion (Law of Inertia) with one suitable sports example. [Practice] | Confidence: Very High

Q25. List down any four factors that affect the trajectory of a projectile. [PYQ] | Confidence: Very High

Q26. What is a Lever? Name its three main components. [Practice] | Confidence: High

Q27. Differentiate between Abduction and Adduction movements with examples. [PYQ 2022] | Confidence: Very High

Q28. State any two methods by which friction can be purposefully reduced in sports. [Practice] | Confidence: High

Q29. Briefly explain the 'Base of Support' in relation to the stability of an athlete. [Practice] | Confidence: High

Q30. Explain the Second-class lever with a suitable sports or anatomical example. [PYQ] | Confidence: High

Q31. Why is friction considered advantageous in certain sports? Give two examples. [Practice] | Confidence: Medium

Q32. Define Projectile and give two examples from track and field events. [PYQ] | Confidence: Very High

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3-Mark

Q33. With the help of a suitable sports example, explain the application of Newton's Third Law of Motion in sports. [PYQ 2024] | Confidence: Very High

Q34. In the examples given below, which of Newton's Laws of Motion will apply? Justify your answer. (a) When a swimmer pushes the water backwards. (b) When a cricket ball and a tennis ball are thrown with the same force, the tennis ball moves with greater acceleration. [PYQ 2024] | Confidence: Very High

Q35. A gymnast maintains a handstand position on the balance beam, then performs a flip. Differentiate between the types of equilibrium shown and explain how they help in performance. [PYQ 2025] | Confidence: Very High

Q36. What is Friction? Differentiate clearly between Sliding friction and Rolling friction. [PYQ] | Confidence: High

Q37. Explain the three classes of levers (Class I, II, and III) with one anatomical or sports example for each. [Practice] | Confidence: Very High

Q38. Explain the concepts of Flexion and Extension with the help of anatomical movements in sports. [PYQ 2022] | Confidence: High

Q39. Discuss the application of Newton's First Law of Motion (Law of Inertia) in sports, providing two distinct examples. [Practice] | Confidence: High

Q40. Elaborate on the aerodynamic factors (like air resistance and spin) affecting a projectile's flight. [Practice] | Confidence: Medium

Q41. Explain any three fundamental principles of stability (e.g., broadening the base of support, lowering the centre of gravity) and their direct application in sports. [Practice] | Confidence: High

Q42. Elucidate how the knowledge of biomechanics helps coaches and athletes in the prevention of sports injuries. [PYQ] | Confidence: High

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5-Mark (Long Answer/Case-Study)

Q43. What do you mean by Equilibrium? Explain how Equilibrium increases with the influence of various factors (Centre of gravity, Base of support, Mass of the body) by giving suitable examples from sports. [PYQ] | Confidence: Very High

Q44. State Newton's Laws of Motion. Explain their practical application in sports events with appropriate, real-world examples for each law. [PYQ 2024] | Confidence: Very High

Q45. Define 'Friction' in sports. Distinguish clearly between Static, Sliding, and Rolling Friction. Furthermore, elaborate on the specific methods used to purposefully increase and decrease friction in various sports. [PYQ] | Confidence: Very High

Q46. What is a lever? Discuss the application of Levers in sports in detail, explaining the position of the fulcrum, effort, and load in all three classes of levers. [PYQ 2025] | Confidence: Very High

Q47. Case Study: During a physical education class, Newton’s Laws of motion were discussed, and their practical application in sports events was explained to students. These laws are most relevant in sports as most of the actions in sports are related to these laws. (i) Newton’s First law of motion is also known as? (ii) Newton’s Second law of motion is also known as? (iii) Newton’s Third law of motion is also known as? (iv) Give one practical example of Newton's Second Law from the sport of Athletics. [PYQ 2022] | Confidence: Very High

Q48. Case Study: Sohan, a new student in the school, was very much interested in taking part in the school sports events. He was taught the latest rules and regulations of the football game. In due course, he learnt the biomechanical aspects of the game including various anatomical movements. (i) The type of movement in which the angle between the joint decreases is called _**. (ii) Straightening parts of a joint so that the angle increases is called _**. (iii) Moving a body part away from the mid-line is called _**. (iv) Moving a body part towards the mid-line is called _**. [PYQ 2022] | Confidence: Very High

Q49. Case Study: Riya, a dedicated track and field athlete, is practicing her shot-put throws. Her coach brings her to the physics lab to explain the biomechanics of projectile motion to maximize her throwing distance. (i) The parabolic path followed by the shot-put in the air after being released is known as a ___. (ii) Assuming the release height and landing height are level, what is the ideal mathematical angle of release for maximum horizontal distance? (iii) Name the invisible force that pulls the shot-put back to the ground. (iv) State any one aerodynamic factor (other than gravity) that actively affects the trajectory of the projectile in the air. [Practice] | Confidence: High

Q50. Define Biomechanics. Discuss comprehensively its importance in improving techniques, designing sports equipment, and promoting safety/injury prevention in sports. [PYQ] | Confidence: Very High

Answer Key

Q1. (C) Newton's Third Law **Q2.** (B) Second-class lever **Q3.** (A) Law of Inertia **Q4.** (C) Buoyant force **Q5.** (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). *(R explains dynamic equilibrium, but does not state WHY A is true).* **Q6.** (A) Flexion **Q7.** (B) Extension **Q8.** (C) Abduction **Q9.** (D) Adduction **Q10.** (A) Trajectory **Q11.** (B) Sliding friction **Q12.** (B) Law of Momentum / Acceleration **Q13.** (C) Third-class lever **Q14.** (B) 45° **Q15.** (C) Gravity **Q16.** (A) Both (A) and (R) are true and (R) is the correct explanation of (A). **Q17.** (A) First-class lever **Q18.** (B) Centre of Gravity **Q19.** (B) Dynamic Equilibrium **Q20.** Biomechanics is the study of the structure, function, and motion of the mechanical aspects of biological systems using the methods of mechanics. **Q21.** Biomechanics is the application of mechanical laws to living structures. Importance: 1. Improves sports technique. 2. Helps in designing safer and more efficient sports equipment. **Q22.** Static equilibrium is when the body is completely at rest (e.g., a handstand). Dynamic equilibrium is maintaining balance while the body is in uniform motion (e.g., cycling). **Q23.** CG is the point where the entire weight of the body acts. A lower CG increases an athlete's stability (e.g., bending knees while wrestling). **Q24.** Law of Inertia states an object remains at rest or in motion unless acted upon by a force. Example: A soccer ball remains stationary on the penalty spot until a player kicks it. **Q25.** 1. Angle of release, 2. Initial velocity of release, 3. Height of release, 4. Gravity. **Q26.** A lever is a rigid rod that rotates around a fixed point to overcome resistance. Components: Fulcrum (Axis), Effort (Force), and Load (Resistance). **Q27.** Abduction: Moving a limb away from the midline (e.g., raising arms to the side). Adduction: Moving a limb toward the midline (e.g., bringing arms back to the sides). **Q28.** 1. Using lubricants or powders (e.g., talcum powder in carrom/gymnastics). 2. Making surfaces smoother (e.g., ice skating rinks). **Q29.** Base of support is the area beneath an object/person that includes every point of contact with the supporting surface. A wider base increases stability. **Q30.** In a second-class lever, the Load is between the Fulcrum and Effort. Example: A push-up (fulcrum at feet, load is body weight, effort at hands) or a calf raise. **Q31.** Advantageous for grip. Examples: 1. Spikes in athletics shoes prevent slipping. 2. Chalk on a gymnast's hands prevents slipping off the bar. **Q32.** An object thrown into the space/air horizontally or at an acute angle under the action of gravity is called a projectile. Examples: Javelin throw, Shot put. **Q33.** Newton's Third Law (Action & Reaction). Example: In swimming, a swimmer pushes the water backward (action), and the water pushes the swimmer forward with an equal and opposite force (reaction). **Q34.** (a) Newton's Third Law (Action-Reaction). The push backward results in a forward thrust. (b) Newton's Second Law (Acceleration). $F = ma$. Since the tennis ball has less mass, the same force causes greater acceleration. **Q35.** Handstand on the beam is **Static Equilibrium** (body is stable at rest). The flip is **Dynamic Equilibrium** (maintaining balance while rotating/moving). Static equilibrium provides a stable base before execution, and dynamic equilibrium ensures successful landing without falling. **Q36.** Friction is the resistance to motion of one object moving relative to another. **Sliding friction:** When two solid surfaces slide over each other (e.g., ice skating). **Rolling friction:** When an object rolls across a surface (e.g., a football rolling on grass). Rolling friction is generally much less than sliding friction. **Q37. Class I:** Fulcrum in middle (e.g., Neck joint looking up and down). **Class II:** Load in middle (e.g., Calf raises - fulcrum at toes, load is body weight, effort at calf muscle). **Class III:** Effort in middle (e.g., Bicep curl - fulcrum at elbow, effort by bicep, load is dumbbell). **Q38.** Flexion decreases the angle between bones (e.g., bending the elbow during a bicep curl). Extension increases the angle (e.g., straightening the arm while throwing a ball). **Q39.** 1. A sprinter resting in the starting blocks will remain at rest until muscle force propels them forward. 2. A barbell on the floor will not move until the weightlifter applies upward force to overcome its inertia of rest. **Q40.** 1. **Air Resistance:** Slows down the horizontal velocity of the projectile. 2. **Spin:** Generates the Magnus effect, creating pressure differences that can cause the ball to lift or drop faster (e.g., topspin or backspin in tennis). **Q41.** 1. **Broader Base:** A wider stance (feet apart) increases stability (e.g., defensive stance in basketball). 2. **Lower CG:** Bending knees lowers the CG, making it harder to be toppled (e.g., wrestling). 3. **Line of Gravity:** Keeping the line of gravity within the base of support prevents falling over. **Q42.** Biomechanics analyzes forces acting on the human body. It helps identify faulty techniques that place undue stress on joints (e.g., incorrect bowling action in cricket causing back injury) and allows correction. It also aids in designing protective gear (helmets, shock-absorbing shoes) that dissipates impact forces. **Q43.** Equilibrium is a state of balance where opposing forces cancel each other out. Factors increasing stability: 1. **Centre of Gravity:** Lowering the CG (bending knees in judo) increases stability. 2. **Base of Support:** Widening the feet spreads the area of support, making the athlete stable. 3. **Mass:** Greater body mass means greater inertia and higher stability (e.g., sumo wrestlers). 4. **Line of Gravity:** Must fall strictly within the base of support. **Q44.** 1. **Law of Inertia:** A body remains at rest or uniform motion unless an external force acts on it. (e.g., A sprinter stays in the blocks until muscular force initiates the sprint). 2. **Law of Acceleration ($F = ma$):** Acceleration is directly proportional to force and inversely to mass. (e.g., A shot put requires more force to accelerate than a baseball). 3. **Law of Reaction:** Every action has an equal and opposite reaction. (e.g., A high jumper pushes down hard on the track to be propelled upward). **Q45.** Friction opposes motion. **Static:** Friction before an object starts moving (highest resistance). **Sliding:** Resistance when sliding (e.g., skiing). **Rolling:** Resistance when rolling (e.g., cycling). **Increasing Friction:** Using spikes in sprinting shoes, applying chalk on hands in weightlifting for better grip. **Decreasing Friction:** Polishing carrom boards, waxing skis, wearing streamlined aerodynamic suits. **Q46.** A lever is a simple machine. **Class 1 (Fulcrum in middle):** Example - Head nodding (Fulcrum: Atlanto-occipital joint). Affords balance. **Class 2 (Load in middle):** Example - Push-ups (Fulcrum: Toes, Load: Body weight at CG, Effort: Hands pushing). Affords mechanical advantage for force. **Class 3 (Effort in middle):** Example - Bicep curl (Fulcrum: Elbow, Effort: Bicep insertion, Load: Weight in hand). Affords speed and range of motion. Class 3 is the most common in the human body. **Q47.** (i) Law of Inertia. (ii) Law of Acceleration / Momentum. (iii) Law of Action and Reaction.

(iv) In shot put, the athlete applies maximum force to the heavy ball to achieve greater acceleration and throwing distance ($F = ma$). **Q48.** (i) Flexion. (ii) Extension. (iii) Abduction. (iv) Adduction. **Q49.** (i) Trajectory (or Parabola). (ii) 45°. (iii) Gravity. (iv) Air resistance (or wind speed/spin). **Q50.** Biomechanics applies the laws of mechanics to human performance. **Improving Technique:** Video analysis corrects flaws (e.g., swimming stroke angles) to maximize efficiency. **Equipment Design:** Develops lighter, aerodynamic bicycles, carbon-fiber vaulting poles, and shock-absorbing running shoes. **Safety:** Evaluates joint stresses to prevent repetitive strain injuries and designs effective protective gear like helmets and shin guards, minimizing impact forces.
